HSIN-LING (JUSTIN) HSU

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EDUCATION

National Chengchi University (NCCU) Sep. 2023 – Expected Graduation: Jun. 2027
B.S. Double Major in MIS and Computer Science *Taipei, Taiwan*

- GPA: 4.11 / 4.30 (3.94 / 4.00) since transfer | Beta Gamma Sigma Honor Society (Top 10% of department).
- Transferred from NCCU Department of Mathematical Sciences after freshman year.

RESEARCH INTERESTS

Trustworthy foundation models (LLMs and VLMs) for high-stakes real-world applications such as healthcare and autonomous driving, spanning hallucination, interpretability, and verifiable model repair.

PUBLICATIONS

Total citations: 84; h-index: 3; i10-index: 2 (Nov 2026). * Equal contribution. † Co-senior author.

[1] **Hsin-Ling Hsu**, Zizheng Wang*, Donghua Zhang*, ..., Chenwei Wu†, Liyue Shen†, "MedAction: Towards Active Multi-turn Clinical Diagnostic LLMs." *Third Conference on Language Modeling (**COLM** 2026)*, San Francisco, USA, 2026. ∼29% acceptance rate. [paper link] [code/data link]

[2] **Hsin-Ling Hsu***, Cong-Tinh Dao*, Luning Wang, Zitao Shuai, ..., Fang-Ming Hung, Chenwei Wu, "MedPlan: A Two-Stage RAG-Based System for Personalized Medical Plan Generation." *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL** 2025): Industry Track*, Vienna, Austria, 2025. ∼25.6% acceptance rate. 30+ citations. [paper link]

[3] Chia-Hsuan Hsu*, Jun-En Ding*, **Hsin-Ling Hsu**, ..., Fang-Ming Hung, "RGPO: Ranking-Guided Preference Optimization for Reliable Clinical Reasoning." *IEEE Transactions on Artificial Intelligence (**IEEE TAI**, Q1 Journal)*, 2026. [paper link]

MANUSCRIPTS

[4] **Hsin-Ling Hsu***, Neeraja Kirtane*, Kuan-Hao Huang, "Not Every Mention Is a Claim: Rethinking Object Hallucination Detection in Vision-Language Models." Under review at *ARR October 2026* (targeting NAACL 2027). [paper link] [data link]

[5] **Hsin-Ling Hsu**, Min-Yu Chen*, Nai-Chia Chen*, Yan-Ru Chen*, Yi-Ling Chang*, Fang Yu, "WARP: Guaranteed Inner-Layer Repair of NLP Transformers." *arXiv preprint*, 2026. [paper link] [code link]

[6] **Hsin-Ling Hsu**, Min-Yu Chen*, Nai-Chia Chen*, Yan-Ru Chen*, Yi-Ling Chang*, Fang Yu, "FORGE: Verification-Gated Behavioral Repair for Generative Language Models." *arXiv preprint*, 2026. [paper link] [code link]

[7] **Hsin-Ling Hsu**, ..., Shi-Sheng Sun, "Reasoning Like an LLM, Running Like an LSTM: Concept Graph Distillation for Explainable Vehicle Trajectory Prediction." In preparation.

PATENTS

Hsin-Ling Hsu, Jun-En Ding, Fang-Ming Hung, "Electronic Device for Generating Personalized Assessment Content and Treatment Plan." Taiwan Utility Model Patent No. M682510, granted May 2026. Assignee: Far Eastern Memorial Hospital. [patent link]

RESEARCH EXPERIENCE

Research Collaborator Feb. 2026 – Present
BioMed-AI Lab, ECE/EECS, University of Michigan (Advisor: Prof. Liyue Shen) *Remote*

- Led **MedAction** [1] (COLM 2026): built a tree-structured distillation pipeline with two KG-grounded consistency metrics to synthesize **MedAction-32K** (32,681 multi-turn diagnostic trajectories from 2,896 PMC cases); the fine-tuned 8B model achieves SOTA among open-source models on MedR-Bench.
- Extended it to **MedAction-RL**: SFT+GRPO with a KG-gated test reward for active diagnostic agents that order tests, interpret results, and decide when to stop, raising 27B accuracy from 0.46 to 0.77 on MedR-Bench. [arXiv]

Research Intern Nov. 2025 – Present
FLAIR Lab, CSE, Texas A&M University (Advisor: Prof. Kuan-Hao Huang) *Remote*

- Research on object hallucination in VLMs from a semantic perspective. Built **NAME-Bench** [4], a 25K-mention benchmark distinguishing asserted from non-assertive object mentions (e.g., negation, hedging, purpose clauses), and showed that current hallucination detectors conflate mentioning an object with asserting its existence. Also studying cross-modal representations in VLMs via sparse autoencoders.

Research Assistant Dec. 2024 – Present
Far Eastern Memorial Hospital (Advisor: Dr. Fang-Ming Hung) *New Taipei, Taiwan*

- Designed **MedPlan** [2] (ACL 2025 Industry Track), a SOAP-aligned two-stage RAG system that generates a clinical assessment and then a patient-specific treatment plan, built on the hospital's extensive real-world EHR data in SOAP format; piloted at the hospital and validated by attending physicians on real patient records (90%+ ICD-10 prediction accuracy).
- Led to a Taiwan utility model patent, the National Innovation Award, and MOUs with clinics and HIS vendors; currently integrating the system with HL7 FHIR for interoperability with hospital information systems.

Undergraduate Capstone Research Sep. 2025 – Present
Departments of MIS and Computer Science, National Chengchi University *Taipei, Taiwan*

- **MIS** (Advisor: Prof. Fang Yu): Proposed **WARP** [5], a provable repair framework for NLP Transformers beyond the final layer, posed as a convex QP with margin, preservation, and certified-robustness guarantees; extended it to generative LLMs as **FORGE** [6], which reduces toxicity and social bias at minimal perplexity cost and stays stable in low-resource settings where fine-tuning fails. Funded by an **NSTC Undergraduate Research Grant**.
- **CS** (Advisor: Prof. Shi-Sheng Sun): Distilling the structure of LLM chain-of-thought into a 69K-parameter concept-graph model for explainable vehicle trajectory prediction [7]; matches the strongest baseline at a fraction of its size, runs ∼3000× faster than the LLM, and supports concept-level intervention.

SELECTED COMPETITIONS & HONORS

National Innovation Award 2025
Project: MedPlan [2]. Co-participating with Far Eastern Memorial Hospital. *Taiwan*

2nd Place, HOTAI MaaS Hackathon, (2/233 teams; ∼0.8%) 2024
AI Travel Itinerary Health Check. Won ***∼$7,800 USD*** *prize.* *Taiwan*

3rd Place, LINE FRESH Campus Competition, (3/165 teams; ∼1.8%) 2024
AI dementia care platform. Built health tracking and multilingual chatbot with LLMs. *Taiwan*

Oral Presentation, NTCIR-18 FinTech Special Session (AI CUP 2024) 2025
MLLM-assisted OCR Text Enhancement for Hybrid Retrieval. [paper link] [code link] *Tokyo, Japan*
Travel grant from Taiwan's Ministry of Education (∼$2,500 USD).

ACADEMIC SERVICE & OTHER ACTIVITIES

- **Reviewer:** KDD 2026 AI4Sci Track, IEEE TCE Journal, NeurIPS Workshop on GenAI for Health (2025, 2026), NeurIPS 2025 Workshop on Efficient Reasoning, Texas NLP Symposium 2026, WMW 2026.
- **Talks:** PyCon TW (academic research sharing); Google Developer Group at NCCU (Generative AI, Databases).
- **Industry:** AI Intern, GoFreight, 2024–25; Part-time AI Engineer, ChainSea, 2023–24; integrated AI/LLMs into company workflows. Part-time Engineer, Academia Sinica, 2024–26: game AI interface and search-tree visualizer.
- **Open Source:** Core contributor to InverseUI-Recorder, LLM-based browser automation (300+ GitHub stars).